FusePlan – Presentation script

Separating Complementarity from Ensemble Effects in Partial Floor Plan Retrieval

How to use it. One section per slide, in the order of the deck. Each section opens with the key message (the one sentence to get across), then the points to say in order, then the few numbers worth memorising. Part 2 collects the technical details and the likely questions, grouped by topic. Every number comes from the paper; the full set is in its tables.

The thread of the talk. Partial floor-plan retrieval is a new task → a plan has two views, image and graph, that fail differently → fusing them improves results, but any fusion of two models improves → ensemble controls separate real information from the ensemble effect → the image adds information only when it is trained on the damage, or when the graph has little left.

Part 1 – Slide by slide

Slide 1 – Title

> **Key message.** FusePlan asks how much of a fusion gain is real information and how much is just the effect of combining two models.

- Authors: Giuseppe Angelillis, Enrico Di Maria, Gabriele Germini (University of Modena and Reggio Emilia).
- Topic: retrieving floor plans from incomplete queries by fusing an image branch and a graph branch.
- The subtitle states the question: when the fusion improves, is it complementarity (new information) or an ensemble effect (two models instead of one)?

Slide 2 – Problem and related work

> **Key message.** No existing method retrieves floor plans from a partial query; and existing fusion results cannot tell new information from an ensemble effect.

1. Motivation.

- An architect rarely starts from a complete plan: a few rooms are sketched, or a fragment is reused, and then complete plans containing it are searched for.
- The task: from a partial plan, find the plan it came from among all 67,405 RPLAN plans.
- Concretely: a plan loses part of its rooms, and the system must rank its intact source as high as possible.

2. The gap in the literature.

- LayoutGMN, LayoutGKN, SSIG and Unit Region Encoding learn or define a similarity between *complete* plans and are evaluated on complete plans.
- Partial plans appear only in *generation*: Graph2Plan and PlanCraft.
- No retrieval method takes partial plans as queries (literature search, not exhaustive).

3. Why fusion, and why it needs care.

- A plan exists as an image and as a graph of rooms, and they fail differently: the image keeps the *shape of the hole* a removed room leaves; a partial graph looks like the complete graph of some smaller plan.
- But any fusion of two models with partly independent errors improves, even if the second model brings nothing new.
- Multimodal methods such as CrossOver and MMFE report only the fused accuracy, so they

cannot separate the two effects.

- Research question: does the image add information to the graph, or does it only act as a second model?
- From on-the-fly sketch retrieval comes the metric: an area under the reciprocal-rank curve over completeness levels.

Numbers to remember. 67,405 plans in the gallery.

Slide 3 – Method

Key message. Two independent branches over the same gallery, a late fusion with one weight, and an ensemble control that measures what the image really adds.

1. Shared setting. Both branches index the same gallery of complete plans and receive the *same* removed rooms, so every system can be compared query by query.

2. Vision branch.

- The plan is rendered as an image and encoded by a *frozen* encoder, PE-Spatial (ViT-B/16), without any training.
- Patch tokens are pooled with GeM; then PCA whitening, estimated on the training split only, gives a 768-d unit vector.
- Optionally, a small head trained on the damage replaces the frozen vector (two variants, see slide 6).

3. Graph branch.

- Each room is a node (room type + room box), each adjacency an edge (relation type). The graph comes from the RPLAN record, not from the image.
- A GATv2 encoder produces a 128-d unit vector.
- Trained with InfoNCE on *asymmetric* pairs: a plan with rooms removed and its complete version. This is exactly the task: map an incomplete plan onto the complete one.
- The checkpoint is selected on retrieval robustness, not on the loss: the validation loss ranks configurations in the opposite order of retrieval.

4. Late fusion.

- The two unit vectors are concatenated with weights $\sqrt{\alpha}$ and $\sqrt{1-\alpha}$.
- The inner product of two fused vectors is then $\alpha \cos_{\text{vision}} + (1-\alpha)\cos_{\text{graph}}$: a weighted sum of the two similarities.
- So one exact FAISS search over the concatenated vectors ranks the whole gallery. α is chosen on validation ($\alpha = 0.4$ with the frozen branch).

5. Ensemble control and complementarity gain (the key idea).

- The graph is also fused with a *second graph model*: a second training of the same network, or a GraphSAGE encoder.
- That second model holds the same information as the graph, so whatever it gains is a pure ensemble effect.
- Complementarity gain: CG = gain(vision + graph) − gain(stronger control).
- CG > 0: the image brings something the graph does not have. CG ≤ 0: the gain is explained by having two models.

Numbers to remember. 768-d vision, 128-d graph, $\alpha = 0.4$.

Slide 4 – Experiments

Key message. The graph trained for incompleteness far exceeds the published baseline, and every fusion improves on it, mostly at heavy damage; whether that gain is real information is the next question.

1. Queries.

- A query is a plan with 25, 50 or 75% of its rooms removed.
- In the image the removed rooms are filled with the background and their walls erased; in the graph only the remaining rooms are kept.

2. Metric.

- For each query: reciprocal rank of the source plan (1 if first, 1/2 if second, ...), averaged over the three levels. This is the AUC.
- AUC = 1 means the source always comes first. Tables show it $\times 100$.

3. Protocol.

- Every choice (configurations, checkpoints, fusion weights) is made on validation.
- Before reading the test split, hypotheses and predicted outcomes were written down; the test split was then read once with every weight fixed.
- Reason: choosing anything on the test set would inflate exactly the numbers being reported.

4. Results (test).

- A room-type histogram finds almost nothing: counting rooms does not solve the task.
- Vision alone: 39.6 frozen, 53.3 with the damage-trained head.
- LayoutGKN, retrained on the same pairs: 61.1. The FusePlan graph (GATv2-comb on the slide, W in the paper): 90.3, about 29 points more.
- Where the gap appears: at 25% removal the two graphs are close; at 75% LayoutGKN drops to 0.22 while GATv2-comb still reaches 0.79. Training and selecting for incompleteness makes the difference.
- Every fusion improves on the graph, up to 94.1, and almost all of the gain comes at 75% removal, the only level where the graph leaves room to improve.
- Open question: a gain of about one point alone does not say whether the image helps or whether any second model would. Slides 5 and 6 answer it.

Numbers to remember. Graph 90.3 vs LayoutGKN 61.1; at 75% removal 0.79 vs 0.22; best fusion 94.1.

Slide 5 – Relative strength and ablation

Key message. The fusion gain is not a property of the image: it depends on how strong the graph is relative to the image.

1. The plot (inverted U).

- Each point: one of the 37 graphs of the configuration search, fused with a vision branch. x axis: AUC of the graph alone; y axis: fusion gain. Orange: frozen vision; green: vision with head; filled points: test.
- Graph much weaker than the image: the image dominates, the graph adds little.
- Branches equally strong: the gain peaks, at about +0.20.
- Graph near the ceiling: little is left to fix, and the gain falls to about +0.01 with the final graph.
- The star is an earlier version of the pipeline, with a weaker graph, which measured +0.17. It lies

on the same curve: the large old gain and the small new one do not contradict each other.

2. Graph ablation.

- Three families compared with the same recipe: GCN, GraphSAGE, GATv2; one factor changed at a time, then the useful changes combined.
- Temperature 0.1 is the strongest single change (52 → 75); the combination reaches 91.
- Combining matters most at heavy damage: at 75% removal 0.81 against 0.38 with the temperature change alone.

3. Vision ablation.

- Eight frozen encoders, 52 configurations, selected on the mean AUC of three damages (room removal, crop, patch).
- Whitening helps every encoder, in all 26 settings. Cause: on synthetic renderings the frozen descriptors collapse into a narrow cone, so a damaged query looks similar to almost any plan; whitening spreads them out.
- Newer or larger encoders do not win: TIPSv2, the best on complete plans in the benchmark, ranks sixth under damage.

4. Negative result worth telling (if time allows).

- The first room-removal rendering erased the dark perimeter walls but kept the lighter interior ones, so each removed room stayed as an empty outlined cell.
- This artifact inflated a frozen encoder from 0.30 to 0.54.
- It was caught thanks to a pre-registered check on other damages; this is why vision configurations are selected on the average of three damages.

Numbers to remember. Peak gain ≈ +0.20 at equal strength; old pipeline +0.17; whitening helps in 26/26 settings.

Slide 6 – Complementarity gain

Key message. With a strong graph, a frozen image encoder adds nothing beyond an ensemble effect; damage-trained heads add real information, but the control still explains about half of their gain.

1. Reading the table. Test split, mean of four replicas, AUC points. Fusion gain: fused AUC minus the better branch. Control gain: the same for a second copy of the graph fused with the graph. CG: the difference.

2. The control. A second training of the same graph, fused with it, already gains +1.63.

3. Frozen vision. Gains +1.25, less than the control: CG −0.38. With a strong graph the frozen image encoder adds no information beyond an ensemble effect, while the raw +1.25 alone would have suggested otherwise.

4. Damage-trained heads.

- Two alternative heads on the frozen encoder, each used in its own fusion (never stacked): the single-damage head, trained on room removal; the multi-damage head, trained on room removal, crop and patch.
- They gain +2.79 and +3.66, exceeding the control by +1.16 and +2.03, positive in all four replicas: they carry information of their own.
- Even here the control explains about half of the gain, so the raw gain overstates the contribution of the image by roughly a factor of two.

5. Where the image helps.

- The CG is zero at 25 and 50% removal and becomes positive only at 75%.
- It is largest on small plans (five rooms or fewer), where even the frozen branch passes the control.
- Cause: with few rooms left, the partial graph is compatible with many plans, while the image still keeps the shape of the hole.
- This breakdown is a post-hoc analysis on one seed: it is a description, not a confirmed result.
- Consistent with slide 5: with graphs only as strong as the image, even the frozen vision branch passes the controls clearly (by 8 to 16 points).

Numbers to remember. Control +1.63; CG −0.38 (frozen), +1.16 and +2.03 (heads); about half of the gain is ensemble effect.

Slide 7 – Conclusions

Key message. Graph trained for incompleteness, complementarity only under two conditions, and a general recommendation: always compare a fusion gain with an ensemble control.

1. **Graph branch.** Trained on partial plans, the graph retrieves a plan from half of its rooms almost as well as from a quarter missing (0.954 vs 0.963 at 50% and 25% removal).
2. **Complementarity.** The image adds information of its own only if it is trained on the damage, or if the graph is no stronger than the image.
3. **Recommendation, beyond floor plans.** Part of any fusion gain comes just from combining two models: compare it with an ensemble control, here a fusion of two graph models, to measure what the second modality really adds.

Limitations. RPLAN only, with synthetic damages; graphs from exact records, not parsed from images; one global fusion weight; the "where it helps" analysis is post hoc on one seed.

Future work. Graphs parsed from images, external datasets, a learned or per-query fusion, a damage the heads never saw in training, human relevance judgements.

Part 2 – Details and likely questions

A. Data and queries

Q: What is RPLAN and how is it split?

A dataset of residential plans; each has a colour rendering (rooms coloured by category) and a record with 13 room types, room boxes and typed adjacencies. The gallery holds the 67,405 plans that have both image and record (48 images lack a record), in the official split: 47,126 train, 10,152 validation, 10,127 test. The gallery always contains every plan, as in a search engine; training and whitening statistics use training plans only. Plans are small: 4 to 8 rooms, 6.8 on average.

Q: How many rooms are removed exactly?

round($f \cdot n$) of the n rooms, rounded to the nearest integer (exact halves to the even number). Example with 7 rooms: $1.75 \to 2$, $3.5 \to 4$, $5.25 \to 5$. Which rooms are removed is drawn with a seed fixed per query and shared by the two branches, so image and graph lose the same rooms. Because rooms differ in size, the removed area is on average 0.29, 0.52, 0.71 of the plan.

Q: What are crop and patch?

Two pixel damages used only for the vision branch: crop blanks one rectangle covering a fraction f of the plan; patch blanks whole ViT patches up to a fraction f. They serve to select vision configurations robustly. Branches and fusions are compared on room removal only, the one damage both branches can receive.

Q: How many queries?

2,000 validation queries for every choice, 2,000 test queries for the final reading, drawn with a fixed seed. Checkpoint and head selection use separate validation queries, disjoint from the evaluation ones.

B. Metric and statistics

Q: What does the AUC measure, and why no 0% level?

Self-recovery: the intact source is in the gallery and the damaged query should retrieve it. Per level the score is 1/rank (0 beyond rank 100); the AUC averages the three levels and then the queries. It follows the m@B metric of on-the-fly sketch retrieval. The 0% level is excluded because about 3% of RPLAN images have a pixel-identical twin, which caps undamaged recovery near 0.97 for every system. Self-recovery needs no relevance labels.

Q: What is a paired bootstrap interval?

All systems see the same queries, so for two systems A and B the per-query difference $\mathrm{AUC}_A(q) - \mathrm{AUC}_B(q)$ is computed; pairing removes the difficulty both systems share. The 2,000 queries are resampled with replacement 10,000 times, the mean difference is recomputed each time, and the 2.5th and 97.5th percentiles give the 95% interval. An interval excluding zero means the difference is not due to which queries were drawn.

Q: The bootstrap ignores training randomness: how is that handled?

The whole pipeline is repeated in four replicas (different seeds for graph training and for the removed rooms). An effect is called above training noise when it has the same sign in all four replicas and its mean exceeds twice its standard deviation across replicas.

Q: Why is the test split read only once, with pre-registered predictions?

Any choice made after looking at test results would inflate them. Hypotheses, predicted outcomes and decision rules were written before each test reading. In the second round, 43 of 45 predictions held. Earlier project stages did read the test split with a weaker graph, but none of those readings selects anything reported.

C. Fusion

Q: What exactly does the fusion formula mean?

$z = [\sqrt{\alpha}\, v; \sqrt{1-\alpha}\, g]$, with v and g unit vectors. Then $\langle z_q, z_p \rangle = \alpha \langle v_q, v_p \rangle + (1-\alpha)\langle g_q, g_p \rangle$: the similarity between query q and gallery plan p is a weighted sum of the vision cosine and the graph cosine. The weights are non-negative and sum to 1, so it is a *convex* combination, and $\|z\| = 1$.

Q: Is FAISS run on the concatenated vectors?

Yes: one exact inner-product search over the concatenated vectors (768+128 = 896-d with the frozen branch). Because of the formula above, that single search ranks the gallery by the weighted sum of the two cosines; there is no need to search the branches separately and merge rankings. At $\alpha = 1$ and $\alpha = 0$ it reproduces the two single branches, which serves as a correctness check.

Q: How is α chosen?

On validation, over $\{0, 0.1, \ldots, 1\}$, with a rule fixed before the runs. Neighbouring weights often score almost the same, so an arg-max would pick among them by chance. Every α whose paired difference from the best has a bootstrap interval containing zero counts as tied; the median of the tied values is taken (of two central values, the one closer to 0.5). The rule avoids choosing on noise.

Q: Is $\alpha = 0.4$ used for every fusion, including the inverted-U curve?

No. 0.4 is the weight of the final graph with the frozen vision branch, the same in all four replicas (0.3 and 0.4 tie). Every other fusion, each control and each graph of the curve, selects its own weight with the same rule (the curve graphs on a coarser grid, step 0.2). The test split always uses the weights fixed on validation.

Q: What is a per-query gate, and why is there none?

A mechanism that would adapt α to each query, for instance to how much of the plan is missing. FusePlan uses one static weight. A hard gate (choosing one branch per query) cannot exceed the per-query oracle, and the static fusion already exceeds that oracle (0.916 vs 0.914 with the frozen branch), because summing similarities can rank the source above both branches on the same query. A soft, learned gate remains future work.

D. Ensemble controls and complementarity gain

Q: Which controls, and which one enters the CG?

Two controls fuse the final graph with a second model holding the same information: a second training of the same configuration (different initialisation and batches) and the best GraphSAGE configuration trained with the same recipe. Each selects its own weight. The CG subtracts the control with the larger gain, a conservative choice; with the final graph that is the second copy (+1.63, against +1.45 for GraphSAGE).

Q: How is the gain defined?

Per query, fused AUC minus the AUC of the branch with the higher mean AUC. The CG is computed on the same queries for both fusions, with a bootstrap interval.

Q: Do gain of the control and CG add up to the fusion gain of one fusion?

Only as a summary: they come from two different fusions, so they describe the gain but do not split one fusion additively.

Q: Why is the frozen CG negative?

With a graph near the ceiling, the frozen image encoder corrects fewer of the graph's errors than a second graph model does. Its gain is equivalent to the GraphSAGE control and slightly below the second copy, negative in all four replicas.

E. Graph branch

Q: What are node and edge features?

19 per node: one-hot of 13 room types plus centre, width, height, area and aspect ratio of the room box (standardised). Edges: undirected, one-hot of 10 RPLAN relation types.

Q: What is the architecture?

Three GATv2 layers, hidden size 256, one attention head, edge features in the attention. Node states are sum-pooled and concatenated with the sum-pooled input features (on the one-hot part this equals the room-type histogram). A linear layer projects to 128-d, then L2 normalisation; about 318k parameters. Every graph encoder trained uses the same 128-d output.

Q: Training details?

Symmetric InfoNCE with in-batch negatives, temperature 0.1, batch 256, learning rate 10^{-3}, 600 epochs, training split only. One view is the complete plan, the other removes a random fraction between 25 and 75% of the rooms.

Q: Why asymmetric pairs?

With symmetric augmented pairs every GNN recovers complete plans but fails on damaged queries (AUC below 0.03). The asymmetric pair is the task itself.

Q: Why select the checkpoint on robustness and not on the loss?

Selecting on complete plans picks early, erratic epochs (0.107 vs 0.457 on one trajectory). The InfoNCE validation loss even ranks configurations in the opposite order of retrieval, because in-batch negatives often share the composition of the anchor. The selected epochs fall at 550–592 of 600, so the graph's robustness is a lower bound.

Q: Is LayoutGKN compared fairly?

It is retrained on the same task, same damaged–complete pairs, whole training split, its own upstream hyperparameters, 300 epochs, and scored with the same protocol; with random weights it scores 0.0001, so training matters. It was trained once, so its value is a lower bound. It stays close at 25% removal and drops at 75%.

Q: Why does the histogram baseline score almost zero?

It scores 0.001: counting room types is not enough. The likely reason is that many plans share the same counts, so the source is lost among them; structure and geometry are needed.

F. Vision branch

Q: Do all vision encoders give 768-d vectors?

No. 768-d is PE-Spatial ViT-B, the selected frozen encoder. Other encoders of the benchmark have their own sizes (for example the ViT-H I-JEPA is larger). The single-damage head outputs 256-d; the multi-damage head keeps 768-d.

Q: Input preprocessing?

The rendering is resized so its longer side has 224 pixels, keeping the aspect ratio, and padded to a square with the white background.

Q: What is GeM, and what is the clamp?

GeM is a generalised mean over the patch tokens, per channel: $\left(\frac{1}{N}\sum_i x_i^p\right)^{1/p}$ with $p = 3$; $p = 1$ is average pooling and large p approaches max pooling, so $p = 3$ emphasises the strongest activations. GeM was designed for CNN features after a ReLU, which are non-negative. ViT tokens after the final LayerNorm are signed, and the implementation clamps every value at 10^{-6} before pooling, so negative components become (almost) zero. In practice the descriptor pools only the positive part

of each channel. This differs from textbook GeM; it is a property of the configuration that won the selection on validation, stated for reproducibility.

Q: What does whitening do?

Mean and covariance come from training plans; each descriptor is centred, rotated onto the principal axes and each axis rescaled to unit variance (all 768 components kept), then L2-normalised. Frozen descriptors of synthetic renderings sit in a narrow cone where every plan looks similar; whitening decorrelates and equalises the dimensions, so differences between plans count. It helps in all 26 settings tested.

Q: Two heads: are they stacked?

No. They are two alternatives, each replacing the frozen vector in its own fusion. The single-damage head (the original head) is an MLP $768 \to 1024 \to 256$ applied to gallery and queries, trained on room removal. The multi-damage head (the second version) acts on queries only: a residual MLP added to the frozen whitened vector, so the gallery stays frozen; trained on room removal, crop and patch. Both use InfoNCE with self-supervised positives, no relevance labels.

Q: Does the single-damage head transfer to other damages?

No: it improves room removal ($+0.136$) but loses on crop (-0.139) and patch (-0.101). A head learns the damage it sees; this motivated the multi-damage head.

G. Scope and limits

Q: Why not evaluate relevance on complete plans?

Per-axis relevance scores (composition, topology, geometry) are computed from the same metadata the graph receives, so they are circular for both branches. They are reported only as a description, never used for selection. Self-recovery has no such problem.

Q: Are the fusion gains with the final graph too small to matter?

They are small in absolute terms (about one to four AUC points) but paired, with intervals excluding zero, and stable across four replicas. The point of the paper is precisely that their size depends on graph strength, and that half of them is an ensemble effect.

Q: What would change with graphs parsed from images?

Not tested. Expectation: parsed graphs would be noisier and the graph branch weaker, which by the inverted U would leave more room for the image. It would also remove the circularity of the relevance scores.

Glossary

Self-recovery	Retrieving the intact source of a damaged query from the full gallery.
AUC	Reciprocal rank of the source averaged over 25/50/75% removal, then over queries.
Ensemble effect	Gain obtained by combining any two models with partly independent errors.
Ensemble control	Fusion of the graph with a second graph model, holding the same information.
Complementarity gain (CG)	Fusion gain minus the gain of the stronger control.
Whitening	Centring, decorrelating and rescaling descriptors with training statistics.
InfoNCE	Contrastive loss: pull the positive pair together, push in-batch negatives apart.
Pre-registration	Writing hypotheses and predictions before reading the test split.
Replica	Full rerun of the pipeline with another seed (four in total).